Basic Properties and Rules for Limits

Suppose the functions f and g both have limits at $x = \infty$.

Constant $$\lim_{x \to \infty} k = k$$

Reciprocal Function $$\lim_{x \to \infty} \frac{1}{x} = 0$$

Constant Multiple $$\lim_{x \to \infty} [k\,f(x)] = k \lim_{x \to \infty} f(x)$$

Sum Rule
$$\lim_{x \to \infty} [f(x) + g(x)] = \lim_{x \to \infty} f(x) + \lim_{x \to \infty} g(x)$$

Difference Rule
$$\lim_{x \to \infty} [f(x) - g(x)] = \lim_{x \to \infty} f(x) - \lim_{x \to \infty} g(x)$$

Product Rule
$$\lim_{x \to \infty} [f(x) g(x)] = \left(\lim_{x \to \infty} f(x) \right) \left(\lim_{x \to \infty} g(x) \right)$$

Quotient Rule
$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = \frac{\lim_{x \to \infty} f(x)}{\lim_{x \to \infty} g(x)}$$ <u>provided</u> $\lim_{x \to \infty} g(x) \neq 0$

Algebraic Power $$\lim_{x \to \infty} [f(x)]^n = \left[\lim_{x \to \infty} f(x) \right]^n$$